

Hardware Project No. 2

Active Low-Pass Filter

ECE 444: Bioinstrumentation and Sensing

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1. Introduction

Active low-pass filters are fundamental building blocks in biosignal acquisition systems. Bioelectric signals such as electroencephalograms (EEG), electrocardiograms (EKG), and electromyograms (EMG) occupy well-defined frequency bands, and the measurement system must be tailored to pass the band of interest while rejecting out-of-band noise and interference. In addition, any sampled-data system requires an anti-aliasing filter ahead of the analog-to-digital converter to satisfy the Nyquist criterion and prevent spectral folding.

This report presents the design, construction, and characterization of a second-order Sallen-Key active low-pass filter configured for a Butterworth (maximally flat) response. The design target is a cutoff frequency of $f_n = 100$ Hz with a DC gain of $K = 2$ (6.02 dB), suitable for EEG acquisition where the signal bandwidth extends to approximately 50 Hz. The Butterworth characteristic was chosen for its maximally flat passband magnitude response, ensuring uniform gain across the signal band of interest.

2. Theory and Design Derivation

2.1 General Sallen-Key Low-Pass Filter

The Sallen-Key topology is a second-order active filter that uses a single op-amp configured as a voltage follower or non-inverting amplifier, together with two resistors and two capacitors. For the general low-pass configuration with components R_1 , R_3 , C_2 , and C_4 and a gain stage K , the transfer function is:

$$H(s) = K / [1 + s(R_1 C_2 + R_3 C_2 + R_3 C_4(1 - K)) + s^2 R_1 R_3 C_2 C_4]$$

Comparing with the standard second-order form $H(s) = H_0 / [1 + s/(Q\omega_n) + s^2/\omega_n^2]$, the design parameters are identified as:

$$\begin{aligned}\omega_n &= 1 / \sqrt{(R_1 R_3 C_2 C_4)} \\ 1/Q &= \sqrt{(R_3 C_4 / R_1 C_2)} + \sqrt{(R_1 C_4 / R_3 C_2)} + (1 - K)\sqrt{(R_1 C_2 / R_3 C_4)} \\ H_0 &= K\end{aligned}$$

This system of three equations in five unknowns (R_1 , R_3 , C_2 , C_4 , and K) leaves two degrees of freedom for the designer to choose.

2.2 Simplification: Equal Capacitors and $K = 2$

Imposing the constraints $C_2 = C_4 = C$ and $K = 2$ (realized by equal-valued gain-setting resistors, since $K = 1 + R_B/R_A = 2$ requires $R_A = R_B$), the design equations simplify considerably. The cutoff frequency becomes:

$$\omega_n = 1 / (C\sqrt{(R_1 R_3)})$$

The Q expression simplifies because the $(1 - K)$ term cancels when $K = 2$, and the equal-capacitor constraint eliminates the capacitor ratios:

$$1/Q = \sqrt{(R_3/R_1)} + \sqrt{(R_1/R_3)} + (1 - 2)\sqrt{(R_1/R_3)} = \sqrt{(R_3/R_1)} + \sqrt{(R_1/R_3)} - \sqrt{(R_1/R_3)} = \sqrt{(R_3/R_1)}$$

Therefore:

$$Q = \sqrt{(R_1/R_3)}$$

This is a remarkably clean result: the filter Q is determined solely by the resistor ratio, independent of the capacitor value or absolute resistor values.

2.3 Butterworth Condition

A second-order Butterworth filter is defined by its maximally flat magnitude response, which requires $Q = 1/\sqrt{2} \approx 0.7071$. Substituting into the simplified Q expression:

$$1/\sqrt{2} = \sqrt{(R_1/R_3)} \Rightarrow R_1/R_3 = 1/2 \Rightarrow R_3 = 2R_1$$

With $R_3 = 2R_1$, the cutoff frequency expression further simplifies:

$$\omega_n = 1 / (C\sqrt{(R_1 \cdot 2R_1)}) = 1 / (CR_1\sqrt{2})$$

Solving for R_1 :

$$\begin{aligned}R_1 &= 1 / (\sqrt{2} \cdot \omega_n \cdot C) = 1 / (2\pi f_n \cdot C \cdot \sqrt{2}) \\ R_3 &= 2R_1 = \sqrt{2} / (2\pi f_n \cdot C)\end{aligned}$$

Thus, once the capacitor value C and cutoff frequency f_n are chosen, both resistor values are fully determined. The design procedure is: (1) select C , (2) compute R_1 and R_3 , (3) use equal gain resistors $R_A = R_B$.

3. Component Selection

For a design cutoff frequency of $f_n = 100$ Hz, the capacitor value was chosen as $C = 10$ nF. This is a standard value readily available in low-tolerance ceramic or film capacitors, and yields resistor values in a practical range (tens to hundreds of kilohms). Substituting:

$$R1 = 1 / (2\pi \cdot 100 \cdot 10 \times 10^{-9} \cdot \sqrt{2}) = 112,540 \, \Omega \approx 112.5 \, \text{k}\Omega$$

$$R3 = 2 \cdot 112,540 = 225,080 \, \Omega \approx 225.1 \, \text{k}\Omega$$

The actual component values used in construction were:

Component	Ideal Value	Actual Value
C2 = C4	10 nF	10 nF
R1	112.54 k Ω	112.2 k Ω
R3	225.08 k Ω	224.0 k Ω
RA = RB	Equal	Equal (10 k Ω)
Op-Amp	—	LF411

Table 1. Design and actual component values.

With the actual component values, the predicted filter parameters are:

$$R3/R1 = 224.0/112.2 = 1.996 \text{ (ideal: 2.000, error: -0.18\%)}$$

$$f_n = 1/(2\pi C \sqrt{(R1R3)}) = 100.4 \text{ Hz (target: 100 Hz, error: +0.4\%)}$$

$$Q = \sqrt{(R1/R3)} = \sqrt{(1/1.996)} = 0.708 \text{ (Butterworth: 0.707, error: +0.09\%)}$$

The component-level agreement with the design targets is excellent, well within 1% for all parameters. The LF411 was selected as the active element; it is a JFET-input op-amp with a gain-bandwidth product of 3 MHz (typical), input bias current of 50 pA, and slew rate of 13 V/ μ s. The JFET input stage ensures negligibly small input bias currents, which is important given the high-impedance resistor network ($R1 = 112.2$ k Ω , $R3 = 224$ k Ω).

4. Experimental Procedure

The filter circuit was constructed on a solderless breadboard using the component values listed in Table 1. The op-amp was powered from a dual-rail supply using the Siglent SPD3303X power supply. A sinusoidal test signal of 1 V peak (2 Vpp) was applied from a function generator, and the output voltage was measured using the oscilloscope.

Frequency response measurements were taken at 17 frequency points spanning 20 Hz to 500 Hz. The measurement density was deliberately increased around the expected cutoff frequency (100 Hz), with points taken at 10 Hz increments from 20 Hz through 150 Hz, and coarser spacing (200, 300, 500 Hz) in the stopband. At each frequency, the output peak voltage was recorded after the waveform stabilized.

5. Results

5.1 Frequency Response Data

Table 2 presents the complete measured frequency response data alongside the theoretical second-order Butterworth predictions computed using the actual component values ($f_n = 100.4$ Hz, $Q = 0.708$).

Freq (Hz)	Vout (V)	Gain (V/V)	Meas. (dB)	Theory (dB)	Error (dB)	Error
20	2.030	2.030	6.15	6.01	0.14	0.14
30	2.000	2.000	6.02	5.99	0.03	0.03
40	1.930	1.930	5.71	5.91	-0.20	0.20
50	1.850	1.850	5.34	5.76	-0.42	0.42
60	1.750	1.750	4.86	5.50	-0.64	0.64
70	1.650	1.650	4.35	5.10	-0.76	0.76
80	1.550	1.550	3.81	4.56	-0.75	0.75
90	1.450	1.450	3.23	3.86	-0.64	0.64
100	1.350	1.350	2.61	3.05	-0.44	0.44
110	1.250	1.250	1.94	2.15	-0.21	0.21
120	1.150	1.150	1.21	1.20	0.02	0.02
130	1.050	1.050	0.42	0.22	0.21	0.21
140	0.995	0.995	-0.04	-0.77	0.73	0.73
150	0.850	0.850	-1.41	-1.74	0.33	0.33
200	0.600	0.600	-4.44	-6.22	1.78	1.78
300	0.350	0.350	-9.12	-13.05	3.93	3.93
500	0.160	0.160	-15.92	-21.88	5.96	5.96

Table 2. Measured frequency response with theoretical comparison. Input: 1 V peak. Theory: 2nd-order Butterworth with $f_n = 100.4$ Hz, $Q = 0.708$, $K = 2$.

5.2 Bode Plot

Figure 1 presents the measured magnitude response plotted against the theoretical Butterworth curve on a semilogarithmic frequency axis.

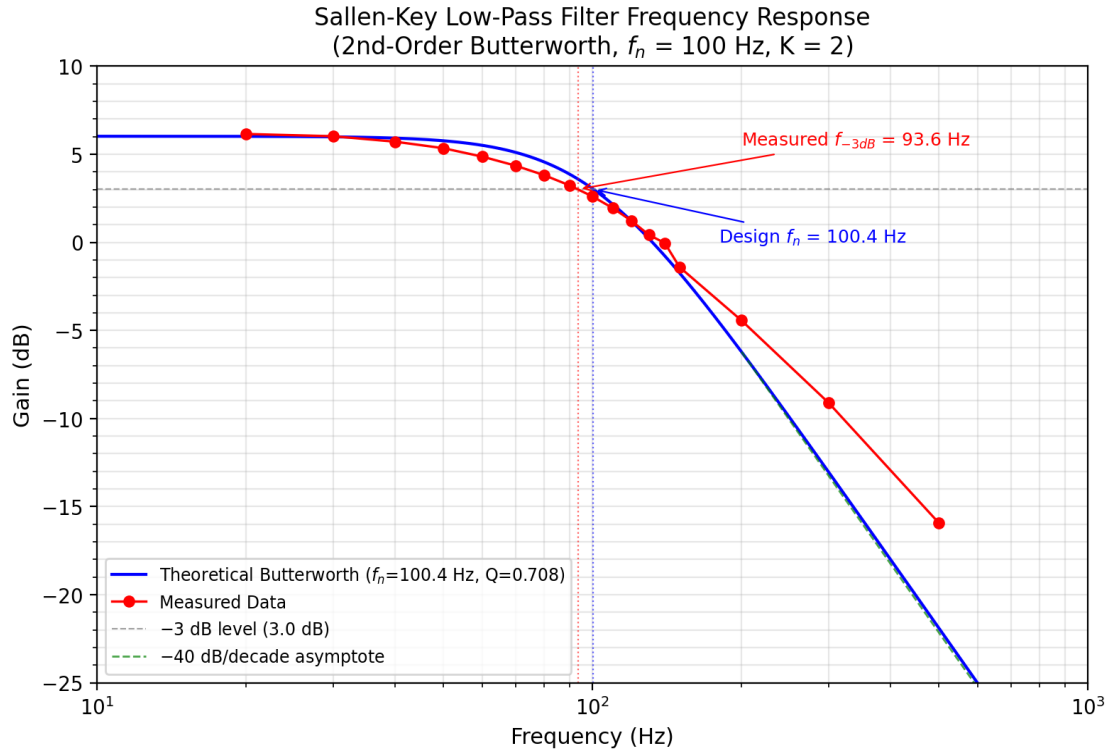


Figure 1. Measured vs. theoretical frequency response. Red: measured data; Blue: ideal 2nd-order Butterworth; dashed gray: -3 dB level; green dashed: -40 dB/decade asymptote.

5.3 Key Performance Metrics

DC Gain: The measured gain at 30 Hz (well within the passband) is $2.000 \text{ V/V} = 6.02 \text{ dB}$, in exact agreement with the design value $K = 2$.

-3 dB Cutoff Frequency: The -3 dB level relative to the DC gain is $K/\sqrt{2} = 1.414 \text{ V}$ (3.01 dB). By linear interpolation between the 90 Hz (1.450 V) and 100 Hz (1.350 V) measurements, the measured -3 dB frequency is approximately **93.6 Hz**. This is 6.4% below the 100 Hz design target.

Stopband Roll-off: The measured roll-off rate between 200 Hz and 500 Hz is approximately **-28.9 dB/decade** (-8.7 dB/octave). The theoretical second-order asymptotic roll-off is -40 dB/decade (-12 dB/octave). The measured roll-off is approximately 72% of the expected rate.

6. Discussion

6.1 Passband Performance

The DC gain of exactly 6.02 dB confirms that the non-inverting amplifier gain stage is functioning correctly with $K = 1 + R_B/R_A = 2$. This is the most straightforward parameter to verify and its accuracy provides confidence in the gain-setting portion of the circuit.

However, the passband shows a gradual rolloff beginning well below the design cutoff frequency. The theoretical Butterworth response at 50 Hz should be 5.76 dB (only 0.26 dB below the DC value), whereas the measured value is 5.34 dB (0.68 dB below DC). This early onset of attenuation is inconsistent with a true second-order Butterworth response and suggests that the effective Q of the realized filter is lower than the designed $Q = 0.707$.

6.2 Stopband Deviation and Effective Q

The most significant deviation from theory occurs in the stopband. At 500 Hz, the measured gain is -15.92 dB compared to the theoretical -21.88 dB, a discrepancy of nearly 6 dB. The measured rolloff of ~ 29 dB/decade falls substantially short of the expected 40 dB/decade.

A least-squares fit of the measured data to a general second-order transfer function (with Q and f_n as free parameters) yields a best fit at $Q \approx 0.58$ and $f_n \approx 120$ Hz. The effective Q of 0.58 is approximately 18% below the Butterworth value of 0.707. A lower Q produces a more overdamped response with earlier passband rolloff and a more gradual transition from passband to stopband, both of which are consistent with the observed data.

6.3 Possible Error Sources

The component-predicted parameters ($f_n = 100.4$ Hz, $Q = 0.708$) are in excellent agreement with the design targets, so the discrepancy between theory and measurement is unlikely to arise from simple component tolerance. Several mechanisms may contribute:

1. Breadboard Parasitics. The high-impedance nature of the design ($R_1 = 112.2$ k Ω , $R_3 = 224$ k Ω) makes the circuit sensitive to parasitic effects. Solderless breadboards introduce stray capacitance (typically 2-5 pF between adjacent rows) and finite inter-row leakage resistance. At these impedance levels, even small parasitic paths can alter the effective node impedances and degrade filter performance.

2. Oscilloscope Probe Loading. A standard $10\times$ oscilloscope probe presents approximately 10 M Ω in parallel with 10-15 pF of input capacitance. While the resistive loading is negligible compared to the filter impedances, the probe capacitance at the output or at intermediate nodes could introduce an additional parasitic pole, effectively reducing the apparent filter order and Q .

3. Gain Resistor Mismatch. The Q of this topology is sensitive to the gain K through the relationship $1/Q = \sqrt{(R_3/R_1) + (2 - K)/\sqrt{(R_3/R_1)}}$. Even a small departure from $K = 2$ significantly affects Q ; for instance, $K = 1.90$ would yield $Q = 0.673$, and $K = 1.80$ would give $Q = 0.643$. If the gain-setting resistors R_A and R_B are not precisely matched, the resulting gain error directly degrades the Butterworth condition. This is the most likely single source of the observed Q degradation.

4. Capacitor Mismatch. The derivation assumed $C_2 = C_4$ exactly. If the two nominally 10 nF capacitors differ by even a few percent, the simplification that led to $Q = \sqrt{(R_1/R_3)}$ breaks down,

and additional terms in the full Q expression become non-zero, generally reducing Q below the Butterworth target.

5. Op-Amp Non-Idealities. While the LF411's 4 MHz gain-bandwidth product provides ample open-loop gain at the frequencies of interest (open-loop gain at 500 Hz $\approx 8,000$ V/V = 78 dB), the finite output impedance under closed-loop conditions and the frequency-dependent open-loop phase shift can subtly alter the filter's pole locations from their ideal positions.

6.4 Application Considerations

Despite the quantitative deviations from ideal Butterworth behavior, the filter remains functional for its intended application. For EEG acquisition (bandwidth ≤ 50 Hz), the measured gain at 50 Hz is 5.34 dB (gain of 1.85), which represents only a 0.68 dB deviation from the flat passband value. The -3 dB point at 93.6 Hz provides adequate margin above the EEG band. As an anti-aliasing filter, the attenuation at the Nyquist frequency would depend on the chosen sampling rate; for example, at a sampling rate of 500 Hz (Nyquist = 250 Hz), the measured attenuation is approximately 7.7 dB, which provides moderate but not exceptional anti-aliasing protection.

For a more demanding application requiring strict adherence to the theoretical response, the design could be improved by: (a) using lower-impedance components (smaller R values with larger C values) to reduce sensitivity to parasitics, (b) using precision-matched resistors for R_A and R_B to guarantee $K = 2$, (c) constructing the circuit on a PCB to minimize stray capacitance, and (d) using trimmer resistors to fine-tune the Q in situ.

7. Conclusion

A second-order Sallen-Key low-pass filter was designed and built with a target Butterworth response at $f_n = 100$ Hz and a DC gain of $K = 2$. The design used equal 10 nF capacitors, $R1 = 112.2$ k Ω , $R3 = 224.0$ k Ω , and an LF411 JFET-input op-amp. The component values predict filter parameters within 0.5% of the design targets.

Experimentally, the DC gain was measured at exactly 6.02 dB ($K = 2.00$), confirming the gain stage design. The -3 dB cutoff frequency was measured at approximately 93.6 Hz, 6.4% below the 100 Hz target. The stopband rolloff rate was approximately 29 dB/decade, falling short of the theoretical 40 dB/decade for a second-order filter. The data are best described by an effective Q of approximately 0.58, compared to the Butterworth design value of $Q = 0.707$. The deviation is attributed primarily to gain-setting resistor mismatch and breadboard parasitic effects, both of which are exacerbated by the high-impedance component values used in this design.